# GLM Revision

## Example 1

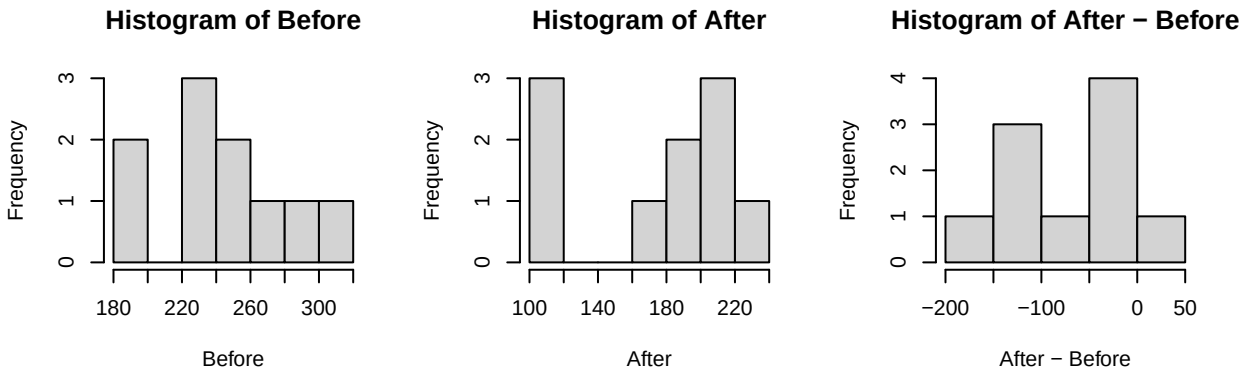
```
Before=c(268,225,252,192,307,228,246,298,231,185)
After=c(106,186,223,110,203,101,211,176,194,203)
Subject=1:10
data1=data.frame(Before,After,Subject)
str(data1)
```

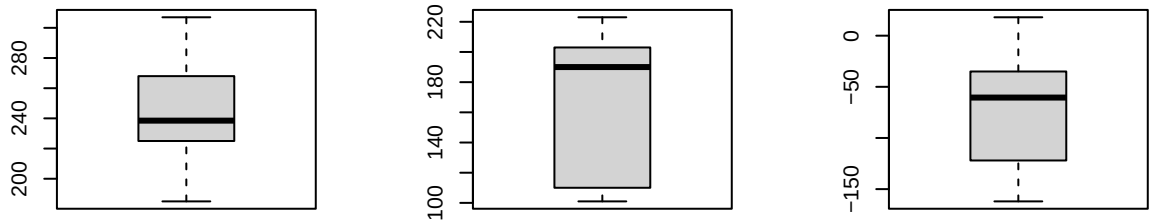
```
## 'data.frame':    10 obs. of  3 variables:
##  $ Before : num  268 225 252 192 307 228 246 298 231 185
##  $ After  : num  106 186 223 110 203 101 211 176 194 203
##  $ Subject: int  1 2 3 4 5 6 7 8 9 10
```

```
data1$Subject=as.factor(data1$Subject)
head(data1)
```

```
##   Before After Subject
## 1    268   106       1
## 2    225   186       2
## 3    252   223       3
## 4    192   110       4
## 5    307   203       5
## 6    228   101       6
```

```
par(mfrow=c(2,3))
hist(Before);hist(After);hist(After-Before)
boxplot(Before);boxplot(After);boxplot(After-Before)
```



```
model1<-lm(After~Before,data1)
head(model1)
```

```
## $coefficients
## (Intercept)      Before
## 120.2155158   0.2100513
##
## $residuals
##          1          2          3          4          5          6          7
## -70.509273  18.522934  49.851548 -50.545372  18.298725 -67.107220  39.111856
##          8          9         10
##  -6.810813  25.262626  43.924988
##
## $effects
## (Intercept)      Before
##  -541.69816    25.28795    67.48880   -47.62839    49.42956   -55.35808
##
##    55.27708    22.11198    37.74778    45.12461
##
## $rank
## [1] 2
##
## $fitted.values
##        1        2        3        4        5        6        7        8
## 176.5093 167.4771 173.1485 160.5454 184.7013 168.1072 171.8881 182.8108
##        9       10
```

```
## 168.7374 159.0750
##
## $assign
## [1] 0 1
```

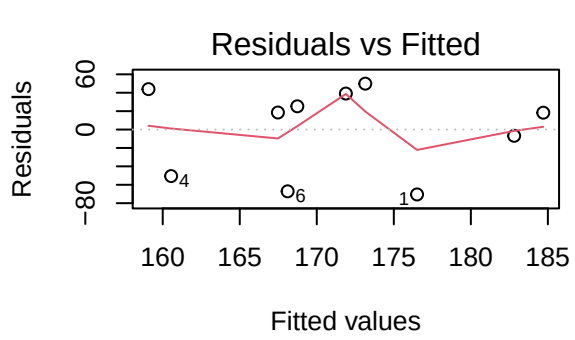
```
model.matrix(model1) #Compute model designed matrix X.
```

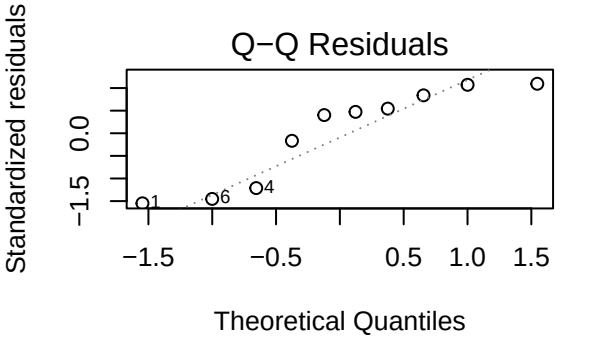
```
##    (Intercept) Before
## 1            1    268
## 2            1    225
## 3            1    252
## 4            1    192
## 5            1    307
## 6            1    228
## 7            1    246
## 8            1    298
## 9            1    231
## 10           1    185
## attr(,"assign")
## [1] 0 1
```

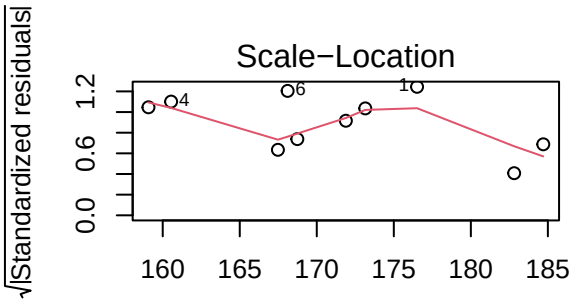
```
summary(model1)
```

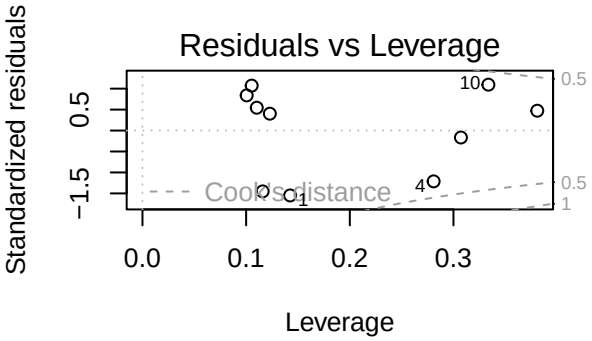
```
##
## Call:
## lm(formula = After ~ Before, data = data1)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -70.51 -39.61  18.41  35.65  49.85
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 120.2155   100.5258   1.196    0.266
## Before        0.2101     0.4084   0.514    0.621
##
## Residual standard error: 49.16 on 8 degrees of freedom
## Multiple R-squared:  0.03201,    Adjusted R-squared:  -0.08899
## F-statistic: 0.2646 on 1 and 8 DF,  p-value: 0.6209
```

```
par(mfrow=c(2,2))
plot(model1)
```




```
data2<-read.csv("/Users/tang/Documents/before_after.csv")
str(data2)
```

```
## 'data.frame':    20 obs. of  3 variables:
##  $ Trt : int  268 225 252 192 307 228 246 298 231 185 ...
##  $ Msr : chr  "Before" "Before" "Before" "Before" ...
##  $ Subj: chr  "subj1" "subj2" "subj3" "subj4" ...
```

```
model1<-lm(Trt~Msr,data=data2)
model.matrix(model1)
```

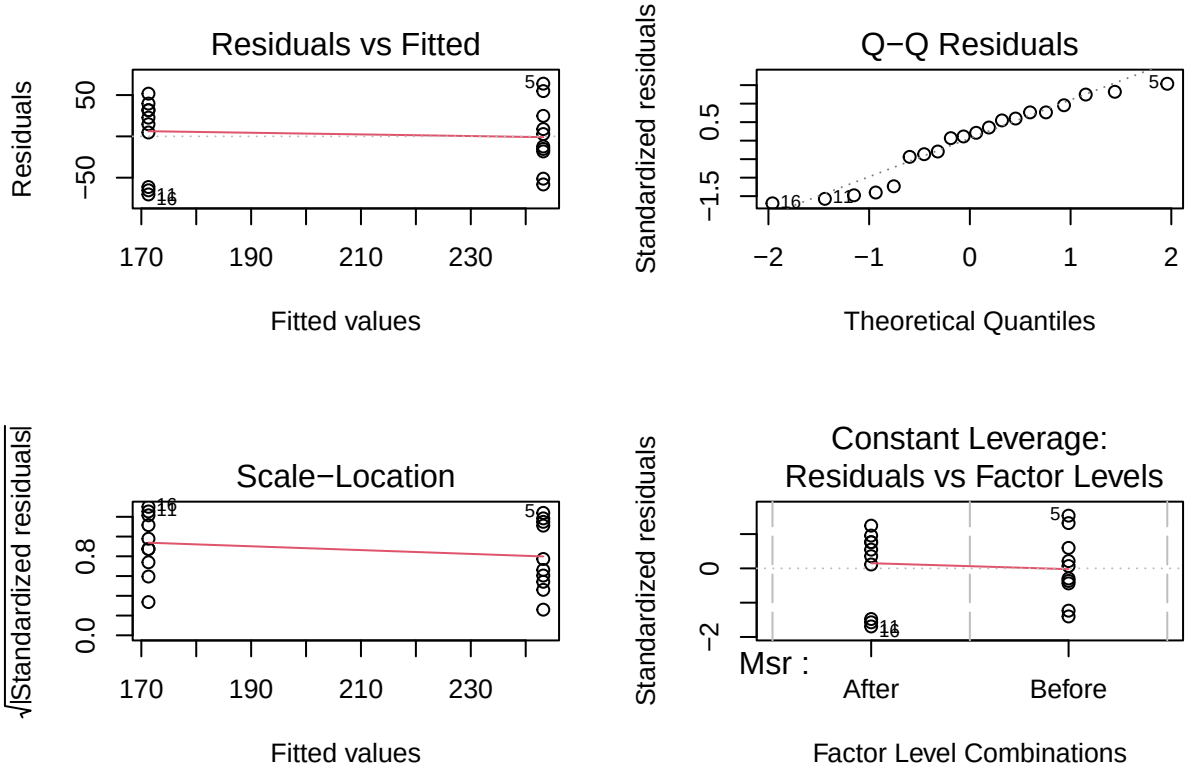
```
##    (Intercept) MsrBefore
## 1            1         1
## 2            1         1
## 3            1         1
## 4            1         1
## 5            1         1
## 6            1         1
## 7            1         1
## 8            1         1
## 9            1         1
## 10           1         1
## 11           1         0
## 12           1         0
## 13           1         0
## 14           1         0
## 15           1         0
```

```
## 16           1         0
## 17           1         0
## 18           1         0
## 19           1         0
## 20           1         0
## attr(,"assign")
## [1] 0 1
## attr(,"contrasts")
## attr(,"contrasts")$Msr
## [1] "contr.treatment"
```

```
summary(model1)
```

```
##
## Call:
## lm(formula = Trt ~ Msr, data = data2)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -70.30 -26.45   6.75  31.70  63.80
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   171.30      13.84  12.379 3.06e-10 ***
## MsrBefore      71.90      19.57   3.674  0.00174 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 43.76 on 18 degrees of freedom
## Multiple R-squared:  0.4285, Adjusted R-squared:  0.3968
## F-statistic:  13.5 on 1 and 18 DF,  p-value: 0.001737
```

```
plot(model1)
```

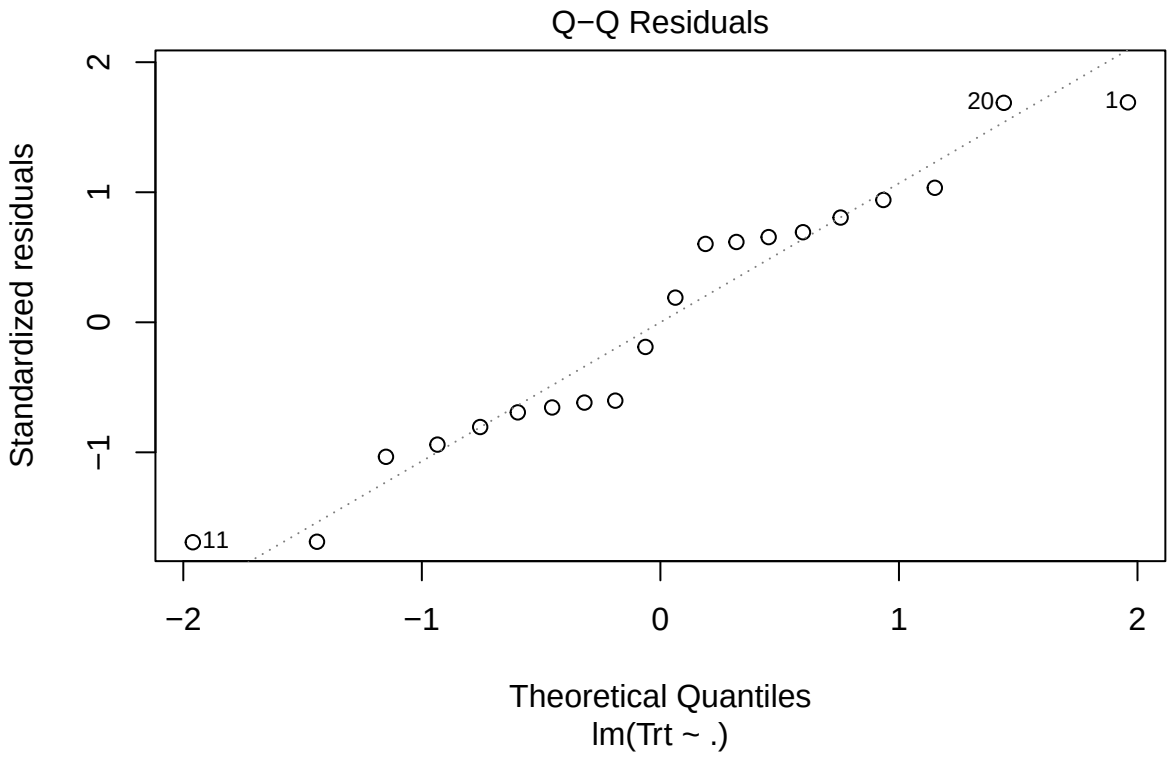



```
data2$Subj<-as.factor(data2$Subj)
data2$Subj<-relevel(data2$Subj,"subj1")
head(data2$Subj);is.factor(data2$Subj)
```

```
## [1] subj1 subj2 subj3 subj4 subj5 subj6
## Levels: subj1 subj10 subj2 subj3 subj4 subj5 subj6 subj7 subj8 subj9
```

```
## [1] TRUE
```

```
model2<-lm(Trt ~ ., data=data2) #Model Trt with all other variables.
par(mfrow=c(1,1))
plot(model2,which=2)
```




### Example 1B

```
var1<-c(12,13,12,42,14,12,32,44,12,4,21,44)
length(var1)
```

```
## [1] 12
```

```
var2<-c("yes","no","yes","yes","yes","no","yes","yes","yes","no","yes","yes")
length(var2)
```

```
## [1] 12
```

```
set<-data.frame(var1,var2)
head(set)
```

```
##   var1 var2
## 1   12  yes
## 2   13   no
## 3   12  yes
## 4   42  yes
## 5   14  yes
## 6   12   no
```

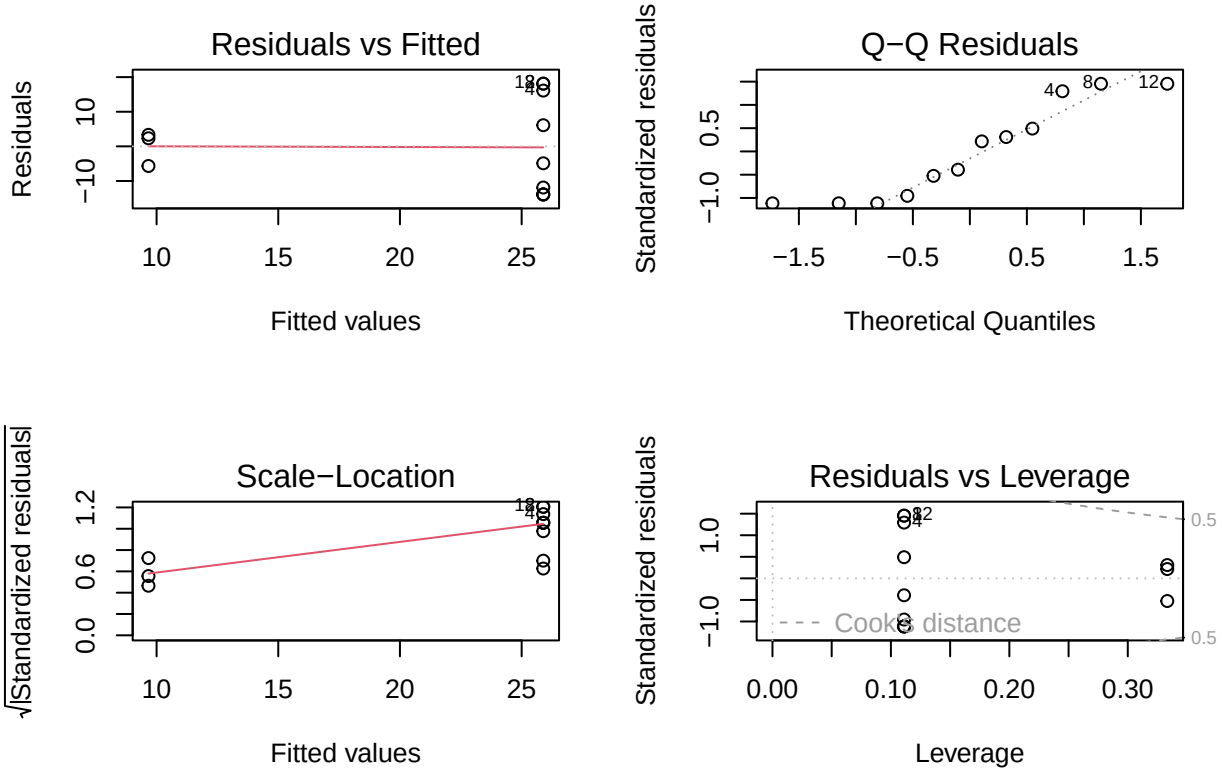
```
set$var2<-as.factor(set$var2) #Making R to treat its values as categories
set$var2<-relevel(set$var2,"no") #chooses one factor category as the reference level.
model1<-lm(var1~var2,data=set)
summary(model1)
```

```
##
## Call:
## lm(formula = var1 ~ var2, data = set)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -13.889 -12.389  -1.278   8.611  18.111
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    9.667      7.628   1.267   0.2338
## var2yes       16.222      8.808   1.842   0.0953 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.21 on 10 degrees of freedom
## Multiple R-squared:  0.2533, Adjusted R-squared:  0.1786
## F-statistic: 3.392 on 1 and 10 DF,  p-value: 0.09532
```

```
model.matrix(model1)
```

```
##    (Intercept) var2yes
## 1            1       1
## 2            1       0
## 3            1       1
## 4            1       1
## 5            1       1
## 6            1       0
## 7            1       1
## 8            1       1
## 9            1       1
## 10           1       0
## 11           1       1
## 12           1       1
## attr(,"assign")
## [1] 0 1
## attr(,"contrasts")
## attr(,"contrasts")$var2
## [1] "contr.treatment"
```

```
par(mfrow=c(2,2))
plot(model1)
```




## Example 2

```
library("GLMsData")
data("turbines")
head(turbines)
```

```
##   Hours Turbines Fissures
## 1   400       39        0
## 2  1000       53        4
## 3  1400       33        2
## 4  1800       73        7
## 5  2200       30        5
## 6  2600       39        9
```

```
names(turbines)
```

```
## [1] "Hours"    "Turbines" "Fissures"
```

```
is.numeric(turbines$Hours)
```

```
## [1] TRUE
```

```
is.numeric(turbines$Turbines)
```

```
## [1] TRUE
```

```
is.numeric(turbines$Fissures)
```

```
## [1] TRUE
```

```
#All variables are quantitative.
summary(turbines$Hours)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     400    1600    2600    2582    3600    4600
```

```
summary(turbines$Turbines)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   13.00   33.50   39.00   39.27   41.00   73.00
```

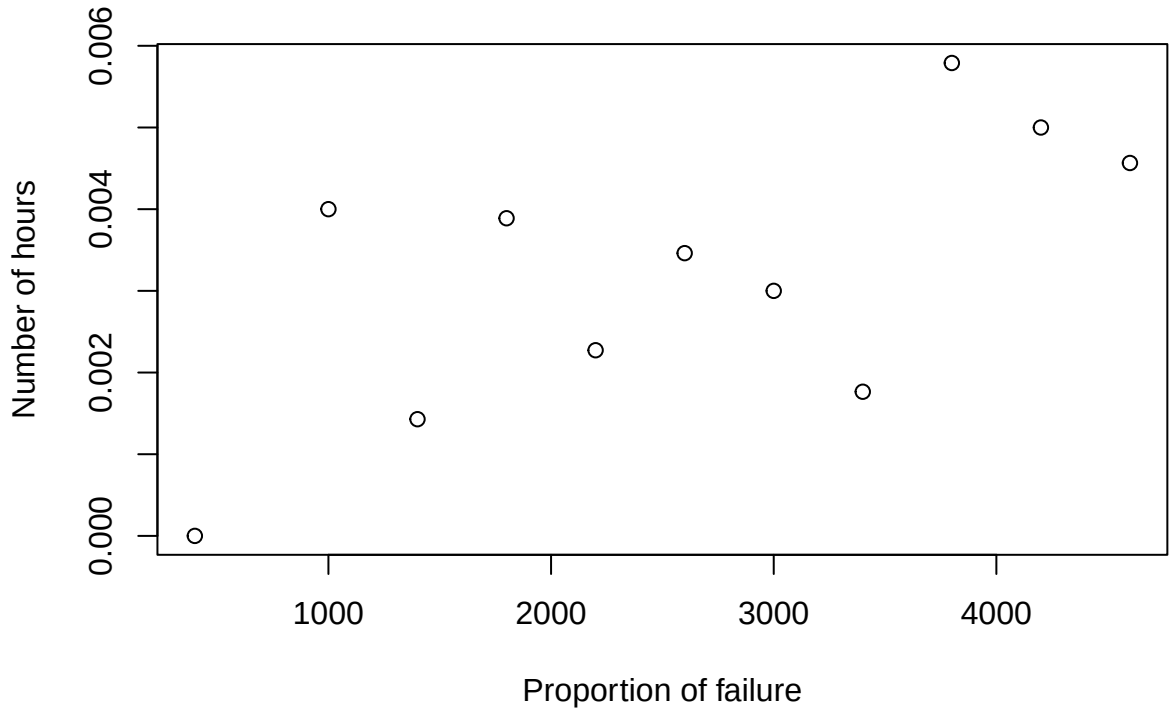
```
summary(turbines$Fissures)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.000   4.500   7.000   9.636  15.000  22.000
```

```
prop<-turbines$Fissures/turbines$Hours
head(prop)
```

```
## [1] 0.000000000 0.004000000 0.001428571 0.003888889 0.002272727 0.003461538
```

```
plot(turbines$Hours,prop,
xlab="Proportion of failure",ylab="Number of hours")
```




```
model1<- lm(turbines$Hours~prop)
summary(model1)
```

```
##
```

```
## Call:
## lm(formula = turbines$Hours ~ prop)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1997.6  -395.4  -118.7   602.4  1560.4
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    925.4      718.2   1.288   0.2297
## prop        518056.0   200056.4   2.590   0.0292 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1083 on 9 degrees of freedom
## Multiple R-squared:  0.427,  Adjusted R-squared:  0.3633
## F-statistic: 6.706 on 1 and 9 DF,  p-value: 0.02924
```

Given that the adjusted R-squared is low (far from 1), the linear regression model does not appear to be appropriate for modeling the data.

## Example 3

```
library("GLMsData")
data("humanfat")
head(humanfat)
```

```
##   Age Percent.Fat Gender  BMI
## 1  23         9.5      M 17.8
## 2  23        27.9      F 22.5
## 3  27         7.8      M 24.6
## 4  27        17.8      M 20.5
## 5  39        31.4      F 25.1
## 6  41        25.9      F 21.4
```

```
names(humanfat)
```

```
## [1] "Age"         "Percent.Fat" "Gender"      "BMI"
```

```
humanfat$Gender<-as.factor(humanfat$Gender)
humanfat$Gender<-relevel(humanfat$Gender,"F")
head(humanfat$Gender)
```

```
## [1] M F M M F F
## Levels: F M
```

```
summary(humanfat$Age)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   23.00   39.50   51.50   46.33   56.75   61.00
```

```
summary(humanfat$Percent.Fat)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    7.80   26.27   30.70   28.61   33.60   42.00
```

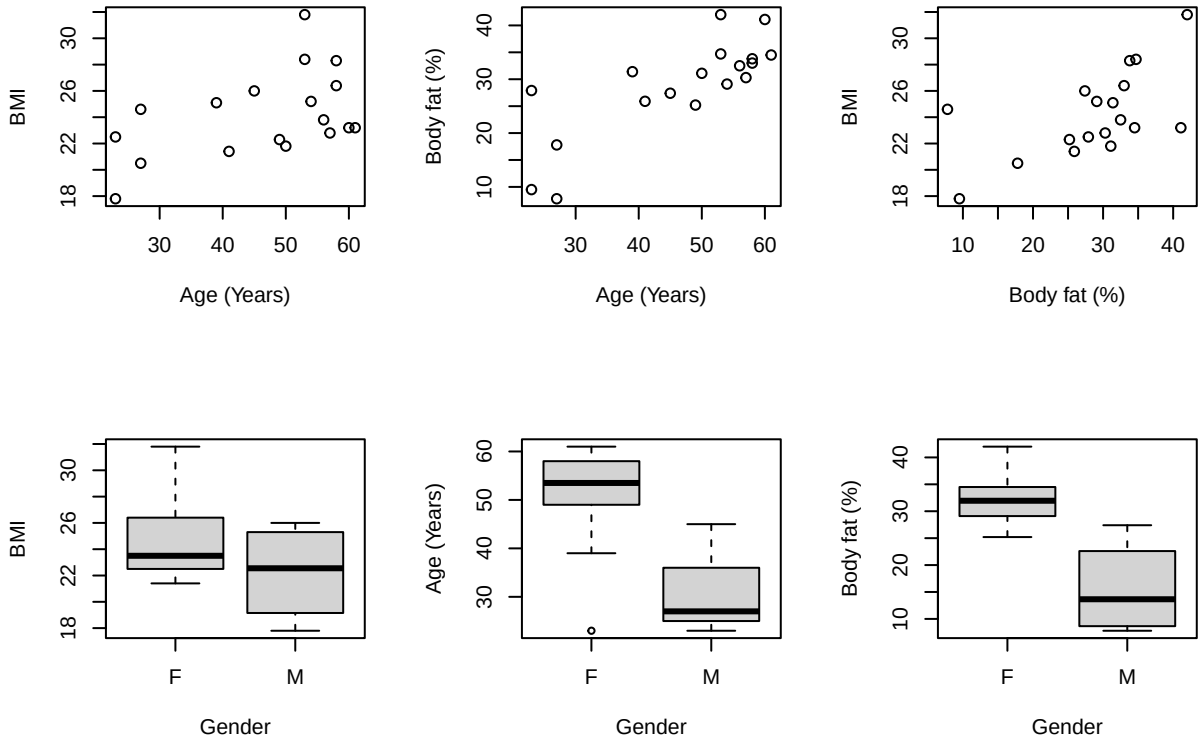
```
summary(humanfat$Gender)
```

```
##  F  M
## 14  4
```

```
summary(humanfat$BMI)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   17.80   22.35   23.50   24.17   25.80   31.80
```

```
par(mfrow=c(2,3))
plot(humanfat$Age,humanfat$BMI,xlab="Age (Years)",ylab="BMI")
plot(humanfat$Age,humanfat$Percent.Fat,xlab="Age (Years)",ylab="Body fat (%)")
plot(humanfat$Percent.Fat,humanfat$BMI,xlab="Body fat (%)",ylab="BMI")
plot(humanfat$Gender,humanfat$BMI,xlab="Gender",ylab="BMI")
plot(humanfat$Gender,humanfat$Age,xlab="Gender",ylab="Age (Years)")
plot(humanfat$Gender,humanfat$Percent.Fat,xlab="Gender",ylab="Body fat (%)")
```




```
model.matrix(~Gender-1,data=humanfat)
```

```
##    GenderF GenderM
## 1        0       1
## 2        1       0
## 3        0       1
## 4        0       1
## 5        1       0
## 6        1       0
## 7        0       1
```

```
## 8        1       0
## 9        1       0
## 10       1       0
## 11       1       0
## 12       1       0
## 13       1       0
## 14       1       0
## 15       1       0
## 16       1       0
## 17       1       0
## 18       1       0
## attr(,"assign")
## [1] 1 1
## attr(,"contrasts")
## attr(,"contrasts")$Gender
## [1] "contr.treatment"
```

```
model1<-lm(BMI~Gender,data=humanfat)
summary(model1)$adj.r.squared
```

```
## [1] 0.05189508
```

```
model2<-lm(BMI~Age,data=humanfat)
summary(model2)$adj.r.squared
```

```
## [1] 0.1900396
```

```
model3<-lm(BMI~Percent.Fat,data=humanfat)
summary(model3)$adj.r.squared
```

```
## [1] 0.3090991
```

```
model1<-lm(BMI~Percent.Fat+Age,data=humanfat)
summary(model1)$adj.r.squared
```

```
## [1] 0.2641485
```

```
model1<-lm(BMI~Percent.Fat+Gender,data=humanfat)
summary(model1)$adj.r.squared
```

```
## [1] 0.3150802
```

```
model1<-lm(BMI~Age+Gender,data=humanfat)
summary(model1)$adj.r.squared
```

```
## [1] 0.1361378
```

```
model1<-lm(BMI~Age+Gender+Percent.Fat,data=humanfat)
summary(model1)$adj.r.squared
```

```
## [1] 0.2697212
```

Observations:

1. BMI is calculated using a person's weight and height. Because muscle mass contributes to body weight, it can influence BMI, although BMI does not distinguish muscle from fat. Therefore, muscle percentage could be included as an explanatory variable when predicting BMI.
2. It is clear that gender is a factor variable.
3. In the first three plots, with BMI as the response variable, the data show considerable scatter. Some observations lie far from the fitted lines, while others are clustered within a narrow range. The plots

alone do not provide strong evidence of a linear relationship between BMI and each quantitative explanatory variable.

4. The adjusted R-squared values for the models using different combinations of body-fat percentage, age, and gender are all below 0.5. This suggests that the models have limited explanatory power in this analysis. However, a low adjusted R-squared alone does not prove that linear regression is unsuitable; residual plots and the models' predictive performance should also be considered.
5. The dataset description states that this is an observational study, not an experiment.
6. The dataset description identifies this as an observational study, not an experiment. Therefore, the results can show associations between BMI and the explanatory variables, but they cannot establish causation. The low adjusted R-squared suggests that the model has limited explanatory power.
7. The sample is small, which limits how confidently the results can be generalized. The participants may not represent the full range of ages and body-fat percentages in the population of interest. Other unmeasured factors, such as diet, physical activity, and muscle mass, may also be related to BMI. Therefore, the findings should be generalized to the broader population with caution.

## Example 4

Recall that the update rule for IRWIS algorithm.

$$\beta^{(r+1)} = (X^T W^{(r)} X)^{-1} X^T W^{(r)} z^r$$

where

$$z_i^{(r)} = \eta_i^r + (y_i - \mu_i^{(r)}) \frac{d\eta_i}{d\mu_i}$$

$$w_{ii} = (Var(Y_i) g'(\mu_i)^2)^{-1}$$

For a poisson GLM with log link:

$$\eta_i = \beta x_i = \log(\mu_i) \implies \mu_i = \exp(\beta x_i)$$

$$\eta_i = g(\mu_i) = \log(\mu_i) \implies g'(\mu_i) = \frac{1}{\mu_i} \implies Var(Y_i) = \mu_i$$

Combining the results, we have

$$z_i^{(r)} = \log(\mu_i^r) + (y_i - \mu_i^r) \frac{1}{\mu_i^r}$$

$$w_{ii} = (Var(Y_i)\, g'(\mu_i)^2)^{-1} = (\mu_i (\frac{1}{\mu_i})^2)^{-1} = \mu_i$$

```
y<-c(0,1)
mu0<-c(0.1,1)
z<-log(mu0)+(y-mu0)*(1/mu0)
w<-matrix(0,nrow=2,ncol=2)
w[1,1]<-mu0[1]
w[2,2]<-mu0[2]

irwis<-function(mu0,x,y,maxn,epsilon){
    beta<-0
    for (i in 1:maxn){
        z<-log(mu0)+(y-mu0)*(1/mu0)
        w<-matrix(0,nrow=length(x),ncol=length(x))
        for (i in 1:length(x)){
```

```
            w[i,i]<-mu0[i]
        }
        beta1<-solve(t(x)%*%w%*%x)%*%t(x)%*%w%*%z
        if (abs(beta-beta1)<epsilon){
            beta<-beta1
            break
        }
        else {
           beta<-beta1
           mu0<-exp(x%*%beta1)
        }
    }
    return(beta)
}
x0<-matrix(c(1,2),nrow=2)
y0<-matrix(c(0,1),nrow=2)
m0<-matrix(c(0.5,0.75),nrow=2)
irwis(m0,x0,y0,100,0.001)
```

```
##            [,1]
## [1,] -0.2474662
```

```
summary(glm(y0~x0-1,family=poisson))
```

```
##
## Call:
## glm(formula = y0 ~ x0 - 1, family = poisson)
##
## Coefficients:
##    Estimate Std. Error z value Pr(>|z|)
## x0  -0.2475     0.5573  -0.444    0.657
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 2.0000  on 2  degrees of freedom
## Residual deviance: 1.7706  on 1  degrees of freedom
## AIC: 5.7706
##
## Number of Fisher Scoring iterations: 5
```

```
x0<-matrix(c(1,2,3,4),nrow=4)
y0<-matrix(c(0,1,3,2),nrow=4)
m0<-matrix(c(0.5,0.75,0.5,0.75),nrow=4)
irwis(m0,x0,y0,100,0.001)
```

```
##           [,1]
## [1,] 0.2071157
```

```
summary(glm(y0~x0-1,family=poisson))
```

```
##
## Call:
## glm(formula = y0 ~ x0 - 1, family = poisson)
##
## Coefficients:
##    Estimate Std. Error z value Pr(>|z|)
```

```
## x0   0.2071     0.1284   1.613    0.107
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 5.3643  on 4  degrees of freedom
## Residual deviance: 3.2830  on 3  degrees of freedom
## AIC: 12.889
##
## Number of Fisher Scoring iterations: 4
```

For example, the GLM for the second case is given by

$$E(Y_i \mid x_i) = \hat{\mu_i} = e^{0.2071 x_i}$$

## Example 4B

For a binomial GLM with logit link:

$$\eta_i = \beta x_i = \log(\frac{\mu_i}{1-\mu_i}) \implies \mu_i = \frac{e^{\eta_i}}{1+e^{\eta_i}} = \frac{e^{\beta x_i}}{1+e^{\beta x_i}}$$

$$\eta_i = g(\mu_i) = \log(\frac{\mu_i}{1-\mu_i}) = \log(\mu_i) - \log(1-\mu_i) \implies g'(\mu_i) = \frac{1}{\mu_i} + \frac{1}{1-\mu_i} = \frac{1}{\mu_i(1-\mu_i)}$$

$$\implies Var(Y_i) = \mu_i(1-\mu_i)$$

Applying these results:

$$w_{ii} = (Var(Y_i) g'(\mu_i)^2)^{-1} = (\mu_i(1-\mu_i)(\frac{1}{\mu_i(1-\mu_i)})^2)^{-1} = \mu_i(1-\mu_i)$$

$$z_i^{(r)} = \log(\frac{\mu_i^{(r)}}{1-\mu_i^{(r)}}) + (y_i - \mu_i^{(r)}) \frac{1}{\mu_i^{(r)}(1-\mu_i^{(r)})}$$

```
irwis<-function(mu0,x,y,n,epsilon){
    beta<-0
    for (i in 1:n){
        w<-matrix(0,nrow=length(x),ncol=length(x))
        z<-log(mu0/(1-mu0))+(y-mu0)/(mu0*(1-mu0))
        for (k in 1:length(x)){
            w[k,k]<-mu0[k]*(1-mu0[k])
        }
        beta1<-solve(t(x)%*%w%*%x)%*%t(x)%*%w%*%z
        if (abs(beta1-beta)<epsilon){
            beta<-beta1
            break
        }
        else{
            beta<-beta1
            mu0<-exp(x%*%beta)/(1+exp(x%*%beta))
        }
    }
    return(beta)
}

x0<-matrix(c(1,2),nrow=2)
```

```
y0<-matrix(c(0,1),nrow=2)
m0<-matrix(c(0.5,0.75),nrow=2)
irwis(m0,x0,y0,100,0.001)
```

```
##           [,1]
## [1,] 0.4196176
```

```
summary(glm(y0~x0-1,family=binomial(link="logit")))
```

```
##
## Call:
## glm(formula = y0 ~ x0 - 1, family = binomial(link = "logit"))
##
## Coefficients:
##    Estimate Std. Error z value Pr(>|z|)
## x0   0.4196     0.9614   0.436    0.662
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2.7726  on 2  degrees of freedom
## Residual deviance: 2.5678  on 1  degrees of freedom
## AIC: 4.5678
##
## Number of Fisher Scoring iterations: 4
```

```
x0<-matrix(c(1,2,3,4),nrow=4)
y0<-matrix(c(0,1,1,1),nrow=4)
m0<-matrix(c(0.5,0.75,0.5,0.75),nrow=4)
irwis(m0,x0,y0,100,0.001)
```

```
##           [,1]
## [1,] 0.8260777
```

```
summary(glm(y0~x0-1,family=binomial(link="logit")))
```

```
##
## Call:
## glm(formula = y0 ~ x0 - 1, family = binomial(link = "logit"))
##
## Coefficients:
##    Estimate Std. Error z value Pr(>|z|)
## x0   0.8261     0.7178   1.151     0.25
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 5.5452  on 4  degrees of freedom
## Residual deviance: 2.9622  on 3  degrees of freedom
## AIC: 4.9622
##
## Number of Fisher Scoring iterations: 5
```

For example, the GLM for the second case is given by

$$\hat{p_i} = \hat{\mu_i} = \frac{e^{0.8261x_i}}{1 + e^{0.8261x_i}}$$

## Example 5

```
data1<-read.csv("/Users/tang/Documents/fish_length(1).csv")
head(data1)
```

```
##   Length River
## 1    135  Wear
## 2    140  Wear
## 3    162  Wear
## 4    136  Wear
## 5    156  Wear
## 6    150  Wear
```

```
data1$River<-as.factor(data1$River)
data1$River<-relevel(data1$River,ref="Wear")

model1<-glm(Length~River,data=data1,family=gaussian)
model2<-glm(Length~River,data=data1,family=Gamma)
model3<-glm(Length~River,data=data1,family=poisson)

summary(model1)
```

```
##
## Call:
## glm(formula = Length ~ River, family = gaussian, data = data1)
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  145.629      1.438 101.239  < 2e-16 ***
## RiverTees     -6.541      1.970  -3.321  0.00113 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for gaussian family taken to be 144.8428)
##
##     Null deviance: 23034  on 149  degrees of freedom
## Residual deviance: 21437  on 148  degrees of freedom
## AIC: 1176
##
## Number of Fisher Scoring iterations: 2
```

```
summary(model2)
```

```
##
## Call:
## glm(formula = Length ~ River, family = Gamma, data = data1)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 6.867e-03  6.941e-05  98.929  < 2e-16 ***
## RiverTees   3.229e-04  9.716e-05   3.324  0.00112 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.007152404)
```

```
##
##     Null deviance: 1.1514  on 149  degrees of freedom
## Residual deviance: 1.0724  on 148  degrees of freedom
## AIC: 1176.6
##
## Number of Fisher Scoring iterations: 4
```

```
summary(model3)
```

```
##
## Call:
## glm(formula = Length ~ River, family = poisson, data = data1)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  4.981059   0.009904 502.916  < 2e-16 ***
## RiverTees   -0.045956   0.013710  -3.352 0.000802 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 162.44  on 149  degrees of freedom
## Residual deviance: 151.21  on 148  degrees of freedom
## AIC: 1174
##
## Number of Fisher Scoring iterations: 3
```

Collecting the results, we have

| | Normal | Gamma | Poisson |
|---|---|---|---|
| $\hat{\beta}_0$ | 145.629 | $6.867 \times 10^{-3}$ | 4.981059 |
| $\hat{\beta}_1$ | -6.541 | $3.229 \times 10^{-4}$ | -0.045956 |
| Null deviance | 23034 | 1.1514 | 162.44 |
| Residual deviance | 21437 | 1.0724 | 151.21 |
| AIC | 1176 | 1176.6 | 1174 |
| Number of iterations | 2 | 4 | 3 |

Table 1: Comparsion between differenet GLMs

The AIC values of all three models are close. However, the Poisson model has the lowest AIC at 1174, compared with 1176 for the Normal model and 1176.6 for the Gamma model. Therefore, the Poisson model is slightly preferred based on AIC. The Gamma model has the smallest null and residual deviances, but these values are based on different distributions and cannot be compared directly. Furthermore, the small p-values for the intercept and rivertees terms provide sufficient evidence that their coefficients are non-zero. Based on these results, the Poisson model is slightly more suitable among the three, although the AIC values do not show a strong difference.

Remarks:

For normal GLM:

$$\eta_i = \beta_0 + \beta_1 x_i = \mu_i$$

For gamma GLM:

$$\eta_i = \beta_0 + \beta_1 x_i = \frac{1}{\mu_i}$$

For poisson GLM:

$$\eta_i = \beta_0 + \beta_1 x_i = \log(\mu_i)$$

# Example 6

Given that

$$\log(\frac{\pi_i}{1-\pi_i}) = -2 + 0.04x_i \implies \pi_i = \frac{e^{-2+0.04x_i}}{1+e^{-2+0.04x_i}}$$

```
generator<-function(n){
    y<-matrix(0,nrow=n,ncol=1)
    x<-runif(n,0,100)
    temp<-exp(-2+0.04*x)
    pi1<-temp/(1+temp)
    for (i in 1:n){
    y[i]<-rbinom(1,1,pi1)
    }
    model<-fitted(glm(y~x,family=binomial))
    return(cor(y-model,model))
}
set.seed(10)
generator(25);generator(100);generator(1000);generator(10000)
```

```
##              [,1]
## [1,] -0.001953977
```

```
##              [,1]
## [1,] 1.722371e-06
```

```
##             [,1]
## [1,] 0.000406967
```

```
##               [,1]
## [1,] -5.927439e-05
```

As $n$ increases, the correlation does not move consistently closer to zero. However, it remains close to zero for all sample sizes, suggesting a weak linear association between the residuals $y_i - \hat{\mu}_i$ and the fitted probabilities $\hat{\mu}_i$. Thus, the centered residual and fitted-probability vectors are approximately orthogonal.

## Part b

Recall that

The LR statistic is

$$LR = 2(l(\beta) - l(\hat{\beta})) \sim x^2_m$$

where $\beta$ is the unrestricted MLE, $\hat{\beta}$ is restricted MLE, and $m$ is the number of restrictions.

Also recall that

The Wald statistic is

$$W = (C\beta - b)^T [C I_n(\beta)^{-1} C^T]^{-1} (C\beta - b) \sim \chi^2_m$$

where $\beta$ is the unrestricted MLE, $C\beta = b$ represents the null hypothesis, $I_n(\beta)$ is the Fisher information matrix evaluated at $\beta$, and $m$ is the number of restrictions.

```
pchisq(c(0.602,14.83,2.83),df=1,lower.tail=FALSE)
```

```
## [1] 0.437815951 0.000117649 0.092518359
```

```r
beta<-c(0.143,1.247,-0.706)
se<-c(0.19,0.45,0.45)
wald<-beta/se
pnorm(-abs(wald),0,1)*2
```

```
## [1] 0.451671336 0.005586536 0.116673831
```

```r
alpha<-0.025
zvalue<-qnorm(1-alpha,0,1)
lower<-beta-zvalue*se
upper<-beta+zvalue*se
cbind(lower,upper)
```

```
##           lower     upper
## [1,] -0.2293932 0.5153932
## [2,]  0.3650162 2.1289838
## [3,] -1.5879838 0.1759838
```

The null hypothesis can be written as

$$H_0 : C\beta = b.$$

For the second explanatory variable, maximum temperature, the hypotheses are

$$H_0 : \beta = 0 \qquad \text{v.s} \qquad H_1 : \beta \neq 0.$$

From the likelihood-ratio test, there is sufficient evidence to reject the null hypothesis that the coefficient for maximum temperature is zero. This indicates that maximum temperature contributes to the model and should be retained.

This conclusion is supported by the Wald test, since the p-value for maximum temperature is below 0.05. Its positive coefficient, $\hat{\beta} = 1.247$, indicates that higher maximum temperatures are associated with higher log-odds of chick survival, holding the other variables constant.

## Part c

Recall, for a normal GLM:

$$\eta_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_n x_{ni} = \mu_i$$

```r
temp<-rnorm(100,1,21)*1.1
temp2<-rbinom(100,1,0.67)
x0<-matrix(temp,nrow=100)
x1<-matrix(temp2,nrow=100)
y<-round(runif(100,0,1),0)
y0<-matrix(y,nrow=100)
model1<-glm(y0~x0+x1,family=binomial)
names(model1)
```

```
##  [1] "coefficients"      "residuals"         "fitted.values"
##  [4] "effects"           "R"                 "rank"
##  [7] "qr"                "family"            "linear.predictors"
## [10] "deviance"          "aic"               "null.deviance"
## [13] "iter"              "weights"           "prior.weights"
## [16] "df.residual"       "df.null"           "y"
## [19] "converged"         "boundary"          "model"
## [22] "call"              "formula"           "terms"
## [25] "data"              "offset"            "control"
## [28] "method"            "contrasts"         "xlevels"
```

```
temp<-summary(model1)
temp<-temp$coefficients
temp
```

```
##                 Estimate  Std. Error    z value  Pr(>|z|)
## (Intercept)  0.313428404 0.301853697  1.0383454 0.2991093
## x0           0.001244352 0.007921394  0.1570875 0.8751759
## x1          -0.140261581 0.409531331 -0.3424929 0.7319800
```

```
estimate<-temp[,1]
se<-temp[,2]
estimate
```

```
##  (Intercept)           x0           x1
##  0.313428404  0.001244352 -0.140261581
```

```
se
```

```
## (Intercept)          x0          x1
## 0.301853697 0.007921394 0.409531331
```

```
wald<-estimate/se
wald
```

```
## (Intercept)          x0          x1
##   1.0383454   0.1570875  -0.3424929
```

```
lower<-estimate-qnorm(0.975,0,1)*se
upper<-estimate+qnorm(0.975,0,1)*se
cbind(lower,upper)
```

```
##                   lower     upper
## (Intercept) -0.27819397 0.9050508
## x0          -0.01428129 0.0167700
## x1          -0.94292824 0.6624051
```

```
library(MASS)
confint(model1)
```

```
## Waiting for profiling to be done...
```

```
##                   2.5 %     97.5 %
## (Intercept) -0.27341464 0.91858056
## x0          -0.01437338 0.01696321
## x1          -0.94954221 0.66193022
```

In this setting, I do not see a noticeable difference between the Wald and profile-likelihood confidence intervals. In other settings, the intervals may differ more visibly. A profile-likelihood interval is based on a likelihood-ratio statistic, which is approximately chi-square distributed under large-sample conditions. A Wald interval uses a normal approximation for the coefficient estimate. With sufficiently large samples and a well-behaved likelihood, the two methods tend to give similar results.

```
wald<-function(design,y){
    model1<-glm(y~design,family=binomial)
    temp<-summary(model1)$coefficients
    estimate<-temp[,1]
    sd<-temp[,2]
    wald<-estimate/sd
    z<-qnorm(0.975,0,1)
    lower<-estimate-z*sd
```

```
    upper<-estimate+z*sd
    print(cbind(lower,upper))
    return(confint(model1))
}
temp<-rnorm(100,1,21)*1.1
temp2<-rbinom(100,1,0.67)
y<-round(runif(100,0,1),0)
y0<-matrix(y,nrow=100)
x<-as.matrix(cbind(temp,temp2))
wald(x,y0)
```

```
##                    lower      upper
## (Intercept) -0.55964925 1.17512995
## designtemp  -0.02712453 0.01235474
## designtemp2 -0.90698909 1.04893442
```

```
## Waiting for profiling to be done...
```

```
##                    2.5 %    97.5 %
## (Intercept) -0.55473969 1.2071148
## designtemp  -0.02771435 0.0122016
## designtemp2 -0.93054496 1.0452937
```

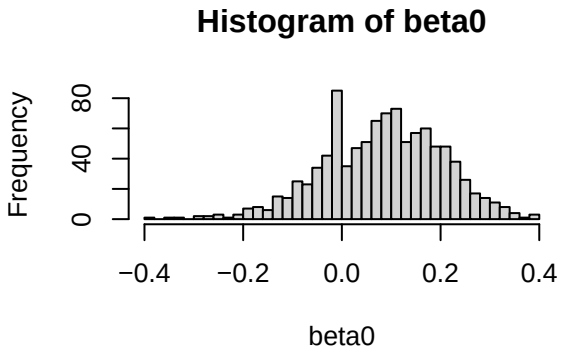
## Part d

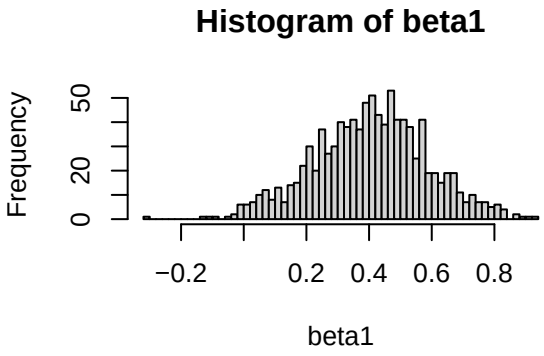
Recall, for a poisson GLM:

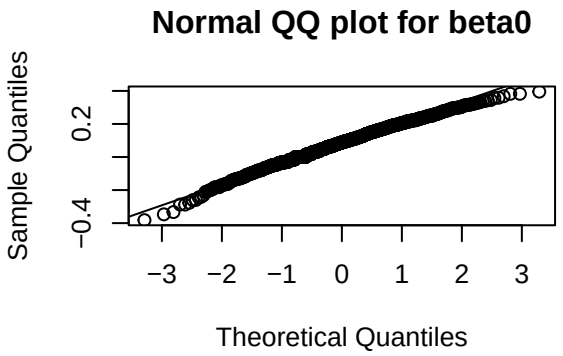
$$\eta_i = \beta_0 + \beta_1 x_i = \log(\mu_i)$$

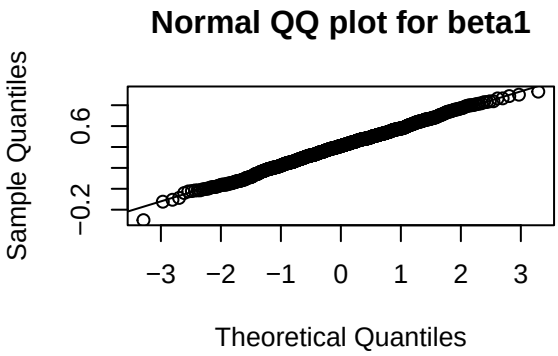
```
simulation<-function(n,r,beta0,beta1){
    set.seed(10)
    temp1<-matrix(0,nrow=n)
    temp15<-matrix(0,nrow=n)
    temp2<-matrix(0,nrow=n)
    temp3<-matrix(0,nrow=n)
    for (i in 1:n){
        x<-rbinom(r,1,0.37)
        eta<-beta0+beta1*x
        mu0=exp(eta)
        y<-rpois(r,lambda=mu0)
        model1<-summary(glm(y~x,family=poisson))
        temp1[i]<-model1$coefficients[1,1]
        temp15[i]<-model1$coefficients[1,2]
        temp2[i]<-model1$coefficients[2,1]
        temp3[i]<-model1$coefficients[2,2]
    }
    z<-(temp2-beta1)/temp3
    z1<-(temp1-beta0)/temp15
    temp4<-cbind(temp1,temp15,z1,temp2,temp3,z)
    colnames(temp4)<-c("estimated beta0","s.d. of beta0","z for beta0",
    "estimated beta1","s.d. of beta1","z for beta1")
    par(mfrow=c(2,2))
    hist(temp1,breaks=50,main="Histogram of beta0",xlab="beta0")
    hist(temp2,breaks=50,main="Histogram of beta1",xlab="beta1")
    qqnorm(temp1,main="Normal QQ plot for beta0")
    qqline(temp1)
```

```
    qqnorm(temp2,main="Normal QQ plot for beta1")
    qqline(temp2)
    return(head(temp4))
}
simulation(1000,100,0.1,0.4)
```




```
##      estimated beta0 s.d. of beta0 z for beta0 estimated beta1 s.d. of beta1
## [1,]      0.12516314     0.1084651  0.23199289       0.3936307     0.1886113
## [2,]      0.09531018     0.1348398 -0.03478068       0.3364722     0.1765468
## [3,]      0.16093037     0.1162474  0.52414382       0.7054885     0.1577247
## [4,]      0.10863384     0.1212660  0.07119752       0.5044706     0.1690986
## [5,]      0.03278982     0.1269991 -0.52921786       0.5958188     0.1716453
## [6,]      0.16332506     0.1170409  0.54105089       0.4757549     0.1660947
##      z for beta1
## [1,] -0.03376972
## [2,] -0.35983521
## [3,]  1.93684590
## [4,]  0.61780893
## [5,]  1.14083453
## [6,]  0.45609467
```

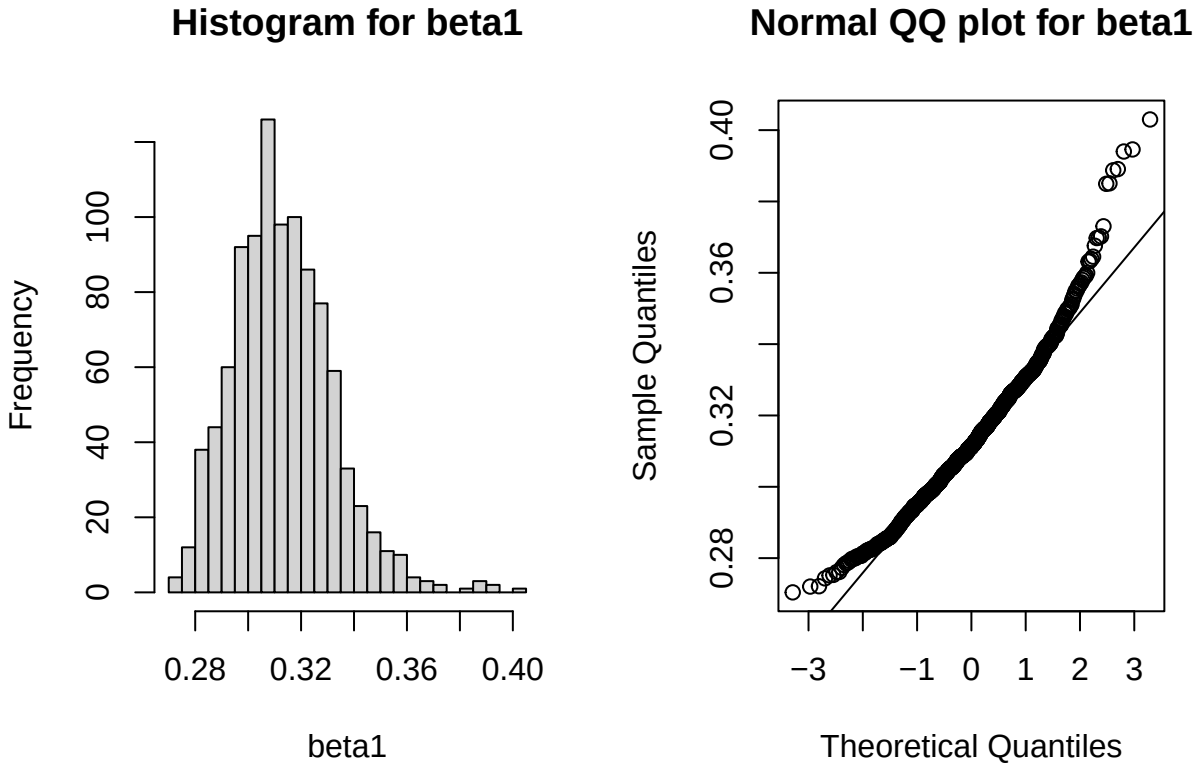
Across 1,000 simulated datasets, each with 100 observations, the histograms of $\widehat{\beta_0}$ and $\widehat{\beta_1}$ appear approximately bell-shaped. Their QQ plots lie reasonably close to the reference lines, supporting the approximate normality of the estimators for this sample size. The 1,000 repetitions help reveal the empirical sampling distributions, while the sample size of 100 in each dataset affects the accuracy of the normal approximation.

**Part dii**

Recall, for a binomial GLM:

$$\eta_i = \beta_0 + \beta_1 x_i = \log\left(\frac{\mu_i}{1-\mu_i}\right) \implies \mu_i = \frac{e^{\eta_i}}{1+e^{\eta_i}} = \frac{1}{1+e^{-(\beta_0+\beta_1 x_i)}}.$$

```
set.seed(1)
simulation<-function(r,lambda,beta0,beta1){
    temp1<-numeric(1000)
    temp2<-numeric(1000)
    temp3<-numeric(1000)
    for (i in 1:1000){
        x<-rpois(r,lambda)
        mu0<-1/(1+exp(-(beta0+beta1*x)))
        y<-rbinom(r,1,mu0)
        model1<-summary(glm(y~x,family=binomial))
        temp1[i]<-model1$coefficients[1,2]
        temp2[i]<-model1$coefficients[2,2]
    }
    par(mfrow=c(1,2))
    hist(temp1,breaks=30,xlab="beta1",main="Histogram for beta1")
    qqnorm(temp1,main="Normal QQ plot for beta1")
    qqline(temp1)
}
simulation(100,1,0,1)
```




Across 1,000 simulated datasets, each with 100 observations, the histogram of $\hat{\beta}_1$ appears approximately

bell-shaped. However, its QQ plot shows a pronounced parabolic pattern. Points deviate substantially from the reference line, especially at theoretical quantiles far from zero. This suggests that the estimator's empirical distribution departs from normality, particularly in the tails. The 1,000 repetitions help reveal the empirical sampling distribution, while the sample size of 100 in each dataset affects the accuracy of the normal approximation.

### Part e

Recall, for gamma GLM:

$$\eta_i = \beta_0 + \beta_1 x_i = \frac{1}{\mu_i}$$

```
set.seed(1)
simulation<-function(x,beta0,beta1,shape,alpha,x0){
    n<-length(x)
    mu=exp(beta0+beta1*x)
    y<-rgamma(n,shape=shape,rate=shape/mu)
    temp<-glm(y~x,family=Gamma(link="log"))
    model<-summary(temp)
    constant<-model$coefficients[1,1]
    estimate<-model$coefficients[2,1]
    sd<-model$coefficients[2,2]
    test<-qnorm((1-alpha/2),0,1)
    lower<-estimate-sd*test
    upper<-estimate+sd*test
    ci<-cbind(lower,upper)
    colnames(ci)<-c("Lower bound","Upper bound")
    mean1<-predict(temp,newdata=data.frame(x=x0),type="response")
    print(paste("Mean response at a given covariate value:", mean1))
    return(ci)
}
x<-rnorm(1000,0,1)
simulation(x,1,2,1,0.05,2)
```

```
## [1] "Mean response at a given covariate value: 176.486759221617"
```

```
##      Lower bound Upper bound
## [1,]    2.014406    2.129596
```

The coefficient intervals are symmetric on the coefficient scale. For the mean response, the interval is calculated on the log scale and then exponentiated, so its bounds are generally asymmetric around the estimated mean on the original scale. This keeps the interval positive, as a Gamma mean must be.

## Question 7

```
beta<-c(-1.2,0.45,0.8)
se<-c(0.25,0.12,0.35)
z<-beta/se
2*pnorm(-abs(z),0,1)
```

```
## [1] 1.586656e-06 1.768346e-04 2.227098e-02
```

Applying the Wald test, all three p-values are less than 0.05. Therefore, we have sufficient evidence to reject the null hypothesis for each coefficient. In the sense that all three coefficients are non-zero.

```r
dfull<-112.4;dreduced<-118.9
testst<- dreduced-dfull
pchisq(testst,df=1,lower.tail=FALSE)
```

```
## [1] 0.01078745
```

Recall that the difference in deviances for nested models follows a chi-squared distribution, with degrees of freedom equal to the difference in the number of parameters. Since the p-value is less than 0.05, we have sufficient evidence to reject the null hypothesis. In this sense, $\beta_2$ is non-zero, which aligns with the results from the Wald test.

## Part b

```r
da<-210.5;db<-175.2;dc<-162.8
dfa<-99;dfb<-98;dfc<-97
test1<-da-db
pchisq(test1,df=dfa-dfb,lower.tail=FALSE)
```

```
## [1] 2.826321e-09
```

```r
test1<-db-dc
pchisq(test1,df=dfb-dfc,lower.tail=FALSE)
```

```
## [1] 0.0004293338
```

Applying the previous methods, both p-values are much smaller than 0.05. Hence, Model B significantly improves the fit compared with Model A.

Applying the previous methods again, the p-value is also much smaller than 0.05, so Model C significantly improves the fit compared with Model B. Therefore, Model C is selected.

## Part c

Given that the full fitted mode is

$$\eta_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_6 x_{6i} = g(\mu_i) = \log(\mu_i)$$

Given that the response variable is

$$Y_i \sim Po(\lambda) = Po(\mu_i)$$

Holding the other covariates constant, Irish legislators are expected to spend $\exp(0.3) \approx 1.35$ times as many days in their constituency as British legislators, or about 35% more.

```r
est<-0.3;se<-0.07
test<-qnorm(0.95,0,1)
est-se*test; est+se*test
```

```
## [1] 0.1848602
```

```
## [1] 0.4151398
```

Hence, the 90% confidence interval for the Nation coefficient is [0.1849, 0.4151].

```r
beta<-c(0.23,0.04,0.06,0.01,0.09,0.05,0.3)
se<-c(0.13,0.04,0.04,0.03,0.06,0.02,0.07)
beta/se
```

```
## [1] 1.7692308 1.0000000 1.5000000 0.3333333 1.5000000 2.5000000 4.2857143
```

```r
2*(1-pnorm(abs(beta/se),0,1))
```

```
## [1] 0.0768553714 0.3173105079 0.1336144025 0.7388826804 0.1336144025
## [6] 0.0124193307 0.0000182153
```

At the 5% significance level, only Geographic proximity and Nation are statistically significant.

Yes, since the coefficient for Nation is statistically significant and non-zero, there is evidence of a difference in the number of days spent in the constituency by British and Irish legislators. Irish legislators are expected to spend more days.

No—the Poisson assumption applies to the **response variable**, not the explanatory variables. Explanatory variables can be integer, categorical, or continuous.

A Poisson GLM may not be fully appropriate because the number of days is bounded, whereas a Poisson distribution has no upper limit and assumes the conditional variance equals the conditional mean. However, it is still reasonably useful because the response is a count, and the model can describe how the expected number of days changes with the explanatory variables.

## Part d

```r
dr<-245.3; df=230.1
dfr<-98;dff<-97
diff<-dr-df
pchisq(diff,df=dfr-dff,lower.tail=FALSE)
```

```
## [1] 9.670342e-05
```

```r
z<-1.9
2*(1-pnorm(abs(z),mean=0,sd=1))
```

```
## [1] 0.05743312
```

Applying the likelihood ratio test, we have sufficient evidence to reject the null hypothesis. Hence, we have evidence that $\beta_2 \neq 0$.

However, applying the Wald test, we do not have sufficient evidence to reject the null hypothesis at the 5% significance level. This does not mean that we can claim $\beta_2 = 0$; it means that the Wald test does not provide sufficient evidence that $\beta_2$ differs from 0.

The tests can give different conclusions in a finite sample. Under regularity conditions, the likelihood ratio and Wald tests are asymptotically equivalent, but the difference alone does not show that the sample size is too small.